

$$h_{ab} = h_{mn} (dx^m)_a (dx^n)_b \quad h_{mn} = \text{diag}(1, 1, 1, 1)$$

$$\Rightarrow \partial_a h_{bc} = 0$$

(Covariance: Σ_{ab}^{ca} cannot appear by itself, must be in form) in any law of physics

• Let ϕ be an isometry, then

$$\phi^* T^a \dots_b \dots$$

will result in the same physical quantities at $\mathcal{O}'$ as $T^a \dots_b \dots$ are in $\mathcal{O}$.

Special Relativity:

• Curves are always timelike

proper time

$$\tau = \int (-\eta_{ab} T^a T^b)^{1/2} d\lambda$$

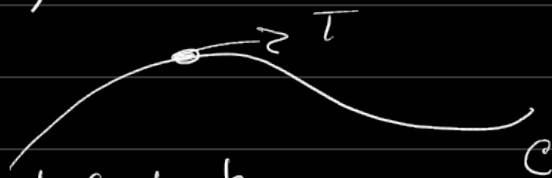
$$\leadsto T^a = dx^a / d\lambda$$

$$\Rightarrow ds^2 = -d\tau^2 = \eta_{ab} dx^a dx^b$$

$$\Rightarrow -\left(\frac{d\tau}{d\lambda}\right)^2 = \eta_{ab} \frac{dx^a}{d\lambda} \frac{dx^b}{d\lambda}$$

$$\left(\frac{d\tau}{d\lambda}\right) = (-\eta_{ab} T^a T^b)^{1/2}$$

$$\therefore \tau = \int (-\eta_{ab} T^a T^b)^{1/2} d\lambda \quad (\text{proper time})$$



τ : time which would elapse on a clock carried along the given curve.

4-velocity: u^a timelike curve parametrized by

$$u^a \equiv \frac{dx^a}{d\tau}$$

$$\Rightarrow -1 = \eta_{ab} u^a u^b$$

• Subject to no external forces u will travel on a geodesic

$$\mathcal{L}(u^a) = 0$$

$$2 \cancel{R_{bc} \partial_a u^b u^c} = 0 \Rightarrow \cancel{R_{bc} \partial_a u^b u^c} = 0$$

$$\partial_a u^c u_c = 0$$

$$\bullet \frac{d}{d\tau} = \frac{dx^a}{d\tau} \partial_a = u^a \partial_a$$

$$\Rightarrow \text{Geodesic: } \frac{d^2 x^a}{d\tau^2} = 0 = \frac{d u^a}{d\tau}$$

also written as

$$u^a \partial_a u^b = \left(\frac{dx^a}{d\tau} \frac{\partial}{\partial x^a} \right) u^b = \left(\frac{d}{d\tau} \right) u^b = 0$$

$$\boxed{\frac{d u^a}{d\tau} = u^a \partial_a u^b = 0}$$

4-momentum: $\boxed{P^a \equiv m u^a}$

Energy: (measured by an external observer present at the site of the particle)

$$\boxed{E = -g(P, V)}$$

$$= -P_a V^a$$

V^a : velocity of the observer relative to $\mathcal{U}$

$$E = -m u_a V^a$$

$$u_a V^a < 0 \Rightarrow$$

$$E = m\alpha$$

$$u_a V^a = -\alpha$$

• For a particle at rest wrt the observer $V^a = u^a$

$$\underline{E = mc^2}$$

• Energy density: $T_{ab} V^a V^b \geq 0$ ("normal" matter)

• Momentum density: $\nabla_\mu (x^a V^a) = 0$
 $-T_{ab} x^a V^b$

• $T_{ab} x^a V^b$:

perfect fluid: $\boxed{T_{ab} = \rho u_a u_b + P (R_{ab} + u_a u_b)}$

comes from asking what is the most general T for a fluid that looks isotropic in its own rest frame.

→ In its own rest frame it is (momentarily) at rest

$$u^a = (1, 0, 0, 0)$$

• In its rest frame it will have

$$T_{ab} = \begin{pmatrix} \rho & 0 \\ 0 & p \end{pmatrix}$$

$$u_a u_b : (1, 0, \dots, 0) : u_a u_b \sim u^a \otimes u_a$$

$$\hookrightarrow \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\Rightarrow u_a u_b \rho = \begin{pmatrix} \rho & 0 \\ 0 & 0 \end{pmatrix} \quad (a)$$

$$N_{ab} P = \begin{pmatrix} -P & 0 \\ 0 & P \end{pmatrix}$$

$$\hookrightarrow N_{ab} P + u_a u_b P \quad (b)$$

$$\therefore T_{ab} = (a) + (b) = u_a u_b \rho + N_{ab} P + u_a u_b P$$

* As a Conserved Quantity

$$\nabla_\mu T^{\mu\nu} = 0 \rightarrow \partial_a T^{ab} = 0 = \partial^a T_{ab}$$

$$\begin{aligned} & \partial^a u_a u_b \rho + u_b \partial^a u_a \rho + u_a u_b \partial^a \rho \\ & + \partial^a P (N_{ab} + u_a u_b) + P (\partial^a u_a u_b + u_a \partial^a u_b) \\ & = (\partial^a u_a) u_b \rho + u_a u_b \partial^a \rho + \partial^a P (N_{ab} + u_a u_b) \\ & + P (\partial^a u_a u_b) = 0 \end{aligned}$$

• $u^b () : (\text{parallel})$

$$\begin{aligned} & = -(\partial^a u_a) \rho - u_a \partial^a \rho + \partial^a P (u_a - u_a) \\ & = P (\partial^a u_a) = 0 \end{aligned}$$

$$\Rightarrow (\partial^a u_a) (\rho + P) + u_a \partial^a \rho = 0$$

(Conservation of Mass)

• ~~with $u \perp u^b : u^a u_a = 0$~~

$$0 = u_a \partial^a u_b u^b \rho + \partial_a P (u_a) + P u_a \partial^a u_b u^b$$

* Note: $0 = u^b [u^a \partial_a (\rho + P) + (\rho + P) \partial^a u_a]$

$$+ (\rho + P) u^a \partial_a u_b + N_{ab} \partial^a P$$

$$(a) [u^a \partial_a (\rho + P) - u_a \partial^a \rho] = u^a \partial_a P$$

$$\Rightarrow 0 = u^b [u^a \partial_a P] + (\rho + P) u^a \partial_a u_b + N_{ab} \partial^a P$$

$$\text{or } (\rho + p) u^a \partial_a u_b + (N_{ab} + u_a u_b) \partial^a P = 0$$

* Non-relativistic limit: $P \ll \rho$, $u^\mu = (1, \vec{v})$
 $v \frac{d\rho}{dt} \ll |\vec{v} P|$

$$\textcircled{a} (\partial_t u_0 + \vec{v} \cdot \vec{v})(\rho + P) + \partial_t \rho + \vec{v} \cdot \vec{\nabla} \rho = 0$$

$$= (\vec{v} \cdot \vec{v})(\rho + P) + \partial_t \rho + \vec{v} \cdot \vec{\nabla} \rho$$

Note: $(\rho + P) \approx \rho$ and

$$\vec{v} \cdot \vec{\nabla} \rho + \vec{v} \cdot \vec{\nabla} \rho = \vec{v} \cdot (\rho \vec{v})$$

since $\partial^i (\rho v_i) = (\partial^i \rho) v_i + \rho \partial^i v_i$
 $= \vec{v} \cdot \vec{\nabla} \rho + \rho \vec{v} \cdot \vec{v}$

and so $\boxed{\frac{\partial \rho}{\partial t} + \vec{v} \cdot (\rho \vec{v}) = 0}$

$$\textcircled{b} \rho u^\alpha \partial_\alpha u_b + (M_{ab} + u_a u_b) \partial^a P = 0$$

$$\bullet \rho \partial_b u_0 + \rho (\vec{v} \cdot \vec{v}) u_b + (-\delta_{0b} + u_b) \partial_t P$$

$$+ (\delta_{ib} + v_i u_b) \partial^i P = 0$$

$b=0$: $v_i \partial^i P = 0$: $\boxed{\vec{v} \cdot \vec{\nabla} P = 0}$

$b=i$: $\rho \partial_t v_i + \rho (\vec{v} \cdot \vec{v}) v_i + (v_i) \partial_t P$
 $+ (\delta_{ij} + v_j v_i) \partial^j P = 0$

$$\rho \partial_t \vec{v} + \rho (\vec{v} \cdot \vec{v}) \vec{v} + \partial_t P \vec{v}$$

$$+ (\vec{\nabla} P + ((\vec{v} \cdot \vec{v}) P) \vec{v}) = 0$$

$\Rightarrow \partial_t P \vec{v} \ll \vec{\nabla} P \Rightarrow (\partial_t P \vec{v}) + \vec{\nabla} P \approx \vec{\nabla} P$

$$\boxed{\rho \left(\frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \vec{v}) \vec{v} \right) = - \vec{\nabla} P}$$

• Note that for a family of $\vec{v}$ observers with parallel 4-vel v^a : $\partial_a v^b = 0$

$T_a = -T_{ab} v^b$ (Mass energy current density
 4-vec of the fluid)

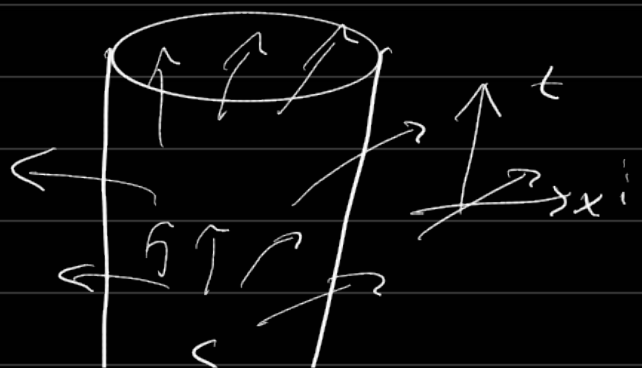
$$\partial^\alpha T_\alpha = 0 : \vec{\nabla} \cdot \vec{J} = 0$$

Gauss' Law: (stokes)

$$\int_S dw = \int_{\partial S} w$$

$$\vec{\nabla} \cdot \vec{J} = 0$$

$$\int_{\partial S} J^\alpha n_\alpha dS = 0$$



* E & B:

$$E_a = F_{ab} v^b$$

$$B_a = -\frac{1}{2} \epsilon_{ab}{}^{cd} F_{cd} v^b$$



